

EBN122 SEMESTER TOETS/TEST 1 (27 AUG 2015)**ANTWOORDBLAD / ANSWER SHEET**

Van / Surname:	Memo	Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language:	Afr		Eng				

Q1		Q2		Tot:	/45
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VRAAG 1 - QUESTION 1 (22)	
1.1	$I_{(4s)} = -2.5 \text{ mA}$ ✓✓
1.2	$I = 20 \text{ mA}$ ✓✓
1.3	$W = 400 \text{ J}$ ✓✓
1.4	$\text{Time} = 3 \text{ hours}$ ✓✓
1.5	$\text{nodes} = 5$ ✓✓ $\text{branches/takke} = 7$ ✓✓
1.6	$P = 6.52 \text{ W}$ ✓✓
1.7	$V_s = 360 \text{ J}$ ✓✓
1.8	$R = 20 \Omega$ ✓✓
1.9	$I_0 = -8.7 \text{ A}$ ✓✓
1.10	$a) R_{eq} = 10 \Omega$ ✓✓ $b) R_{eq} = 10 \Omega$ ✓✓

VRAAG 2 - QUESTION 2 (23)				
2.1	$a = 1$ ✓	$b = -1$ ✓	$c = 3$ ✓	$d = 15$ ✓
2.2	$I_a = 2.5 \text{ A}$ ✓✓	$V_b = 37.5 \text{ V}$ ✓✓		
2.3	$e = -20$ ✓	$f = -10$ ✓	$g = -20$ ✓	$h = 22$ ✓
2.4	$\begin{bmatrix} 1.25 & -0.25 & -0.5 \\ -0.25 & 1.75 & -0.5 \\ -0.5 & -0.5 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ -3 \end{bmatrix}$ ✓✓✓			
2.5	$i_2 = -0.67$ ✓✓	$V_2 = -1$ ✓✓		
2.6	$I_B = 17.55 \mu\text{A}$ ✓✓	$v_0 = 2.525 \text{ V}$ ✓✓		